MODULE 1 · LECTURE 3

Memory Concepts and Units of Memory

How computers represent and store data, the memory hierarchy, RAM, ROM, cache, storage devices and the units used to measure them — with farm-data examples.

Learning objectives

After this lecture you should be able to:

1. Explain how computers represent numbers and text in binary.
2. Convert between bits, bytes and larger units, and tell decimal (SI) prefixes from binary (IEC) prefixes.
3. Describe the memory hierarchy and the trade-off between speed, cost and capacity.
4. Compare the types of RAM and ROM, and explain cache and virtual memory.
5. Describe hard disks, solid-state drives, optical discs, memory cards and cloud storage.
6. Estimate the storage needed for agricultural data and plan backups.

Scope: Lectures 1 and 2 showed where memory sits in a computer. This lecture explains what memory stores, how it is measured and the kinds of memory and storage. Operating-system memory management continues in Lecture 4 and databases in Lecture 6.

1. What is memory?

Memory is the part of a computer that holds data and instructions. It is organised as a long series of **cells** (locations), each with a unique **address**, like numbered lockers. The CPU uses the address to read from or write to a cell.

Term	Meaning
Capacity	How much data the memory can hold (Section 3)
Access time	Time taken to read or write one item
Volatile	Loses its contents when power is switched off (RAM)
Non-volatile	Keeps its contents without power (ROM, flash, disks)
Random access	Any location can be reached directly in about the same time (RAM)
Sequential access	Data must be read in order from the start (magnetic tape)
Direct access	The device moves to the right area, then reads (hard disk)

2. How computers represent data

Electronic circuits are most reliable when they distinguish only two states — on/off, high/low voltage, magnetised one way or the other. Computers therefore use the **binary** number system, with the two digits **0** and **1**. One binary digit is a **bit**.

- With n bits there are $\mathbf{2^n}$ different patterns: 1 bit → 2, 4 bits → 16, 8 bits → 256 (the numbers 0 to 255).
- A group of **4 bits** is a **nibble**; a group of **8 bits** is a **byte** — enough to store one English character.

- A **word** is the number of bits the CPU handles as one unit (32 or 64 bits in modern computers, Lecture 2).

> **Worked example — decimal to binary**
> Convert 45 to binary by dividing by 2 and noting the remainders:
> 45 ÷ 2 = 22 r 1 · 22 ÷ 2 = 11 r 0 · 11 ÷ 2 = 5 r 1 · 5 ÷ 2 = 2 r 1 · 2 ÷ 2 = 1 r 0 · 1 ÷ 2 = 0 r 1
> Reading the remainders from the last to the first: 45 = **101101_2**.
> Check: 32 + 8 + 4 + 1 = 45.
> **Hexadecimal** (base 16) writes each group of 4 bits as one digit 0–9, A–F: $1011\ 0110_2$ = **$B6_{16}$** = 11 × 16 + 6 = 182.

Text: character codes

Code	Size	Covers
ASCII (USA Standard Code for Information Interchange, 1968)	7 bits → 128 characters	English letters, digits, punctuation and control codes
ISCII (IS 13194:1991, Bureau of Indian Standards)	8 bits	Indian scripts in a common layout
Unicode (UTF-8 encoding)	1 to 4 bytes per character	Almost all the world's scripts. Devanagari occupies code points U+0900–U+097F (Unicode Consortium, n.d.)

> **Worked example — storage for text in two languages (UTF-8)**
> "Ram" in English = 3 ASCII characters = 3 bytes.
> "राम" in Devanagari = र (U+0930) + ा (U+093E) + म (U+092E) = 3 code points; each needs 3 bytes in UTF-8, so 9 bytes.
> Farm advisories in Indian languages therefore take more storage and data than the same message in English.

3. Units of memory

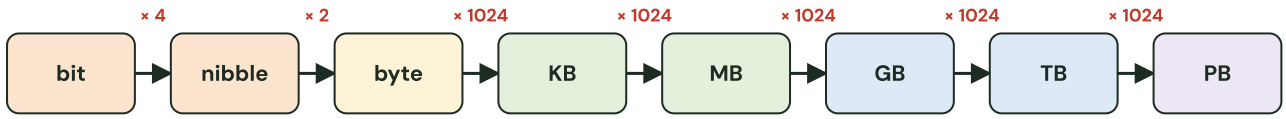


Figure 1. The ladder of memory units (traditional binary convention). Original diagram.

Two conventions are in use. Computer textbooks and most examinations traditionally take **1 KB = 1024 bytes** (2^{10}), because memory chips come in powers of two. Strict SI usage takes **1 kB = 1000 bytes** (10^3), and storage makers label drives this way. To remove the confusion, the International Electrotechnical Commission (IEC) approved separate **binary prefixes** in December 1998 — kibi (Ki), mebi (Mi), gibi (Gi) and so on (National Institute of Standards and Technology [NIST], n.d.).

Decimal (SI) name	Symbol	Bytes	Binary (IEC) name	Symbol	Bytes
kilobyte	kB	10^3 = 1,000	kibibyte	KiB	2^{10} = 1,024
megabyte	MB	10^6	mebibyte	MiB	2^{20} = 1,048,576
gigabyte	GB	10^9	gibibyte	GiB	2^{30} = 1,073,741,824
terabyte	TB	10^{12}	tebibyte	TiB	2^{40}
petabyte	PB	10^{15}	pebibyte	PiB	2^{50}
exabyte	EB	10^{18}	exbibyte	EiB	2^{60}
zettabyte	ZB	10^{21}	zebibyte	ZiB	2^{70}
yottabyte	YB	10^{24}	yobibyte	YiB	2^{80}
ronnabyte	RB	10^{27}	—	—	—
quettabyte	QB	10^{30}	—	—	—

Ronna (R, 10^{27}) and quetta (Q, 10^{30}) are the newest SI prefixes, adopted by the 27th General Conference on Weights and Measures in 2022, partly because data science needs to express quantities of digital information beyond 10^{24} (Bureau International des Poids et Mesures [BIPM], 2022).

Exam tip: unless a question says otherwise, Indian university and competitive examinations usually expect 1 KB = 1024 bytes, 1 MB = 1024 KB, and so on. State the convention you use.

Worked example — why a "1 TB" drive shows about 931 GB
The maker means 1 TB = 10^{12} bytes.
The operating system divides by 2^{30} = 1,073,741,824 bytes per "GB" (really GiB): $10^{12} \div 1{,}073{,}741{,}824 = 931.3$.
So the same drive is reported as about 931 GB. No space is missing — only the unit differs.

Worked example — conversions
(a) 2 MB (binary) in bits = 2 × 1024 × 1024 × 8 = 16,777,216 bits.
(b) 5,000 soil-test records of 200 bytes each = 1,000,000 bytes = 1 MB (decimal) ≈ 0.95 MiB.
(c) 4 GiB = 4 × 1,073,741,824 = 4,294,967,296 bytes — the memory limit of 32-bit addressing seen in Lecture 2.

4. The memory hierarchy

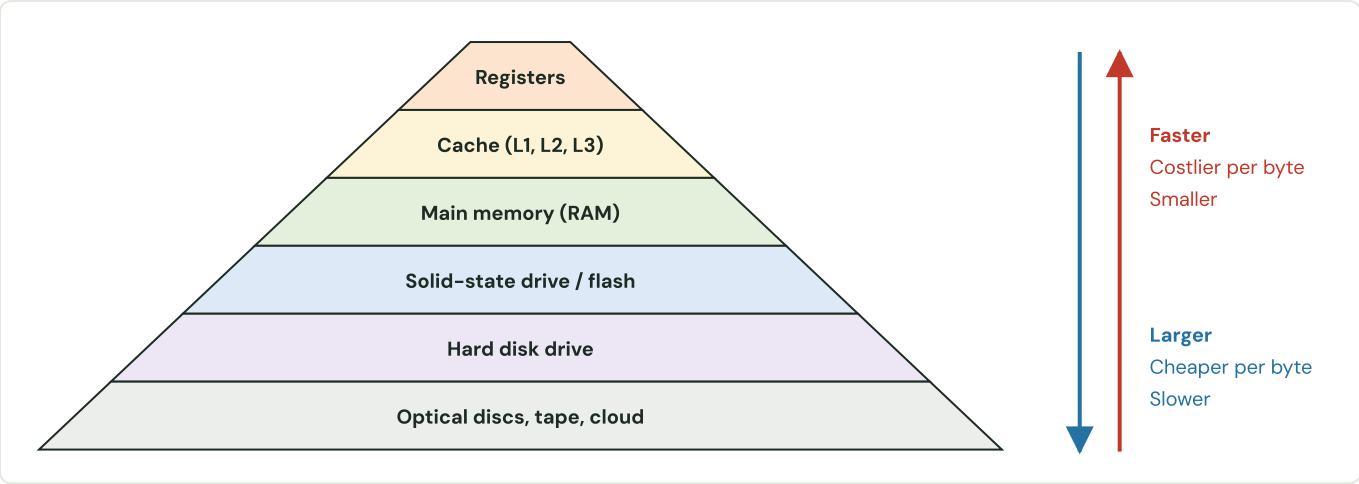


Figure 2. The memory hierarchy: registers, cache and RAM are primary memory; the lower levels are secondary storage. Speed and cost rise towards the top, capacity towards the bottom. Original diagram.

No single memory technology is fast, large and cheap at once. Computers therefore use a **hierarchy**: a little very fast, expensive memory close to the CPU, backed by larger, slower and cheaper layers. Typical access times quoted by Patterson and Hennessy (2014) show the spread:

Level	Technology	Typical access time	Volatile?
Registers	Flip-flops inside the CPU	Under 1 ns	Yes
Cache	Static RAM (SRAM)	About 0.5–2.5 ns	Yes
Main memory	Dynamic RAM (DRAM)	About 50–70 ns	Yes
Solid-state storage	Flash memory	About 5–50 µs	No
Hard disk	Magnetic platters	About 5–20 ms	No
Archive	Optical discs, magnetic tape, cloud	Milliseconds to minutes	No

5. Primary memory

RAM (random-access memory)

RAM holds the operating system, programs and data while they are being used. It can be read and written, and it is **volatile** — unsaved work is lost if the power fails, which is why a UPS matters in offices with unreliable supply.

	Static RAM (SRAM)	Dynamic RAM (DRAM)
Stores each bit in	A flip-flop of several transistors	A tiny capacitor and one transistor
Refreshing	Not needed while powered	Must be refreshed every few milliseconds, because the charge leaks
Speed	Faster	Slower
Cost and density	Costlier; fewer bits per chip	Cheaper; many more bits per chip
Used for	Cache memory	Main memory (RAM modules)

Today's main memory is **DDR SDRAM** (double data rate synchronous DRAM), which transfers data on both edges of the clock signal. Successive generations (DDR3, DDR4, DDR5) are faster and use less power; JEDEC published the DDR5 standard (JESD79-5) in July 2020 (JEDEC, 2020).

ROM (read-only memory)

ROM is **non-volatile**: it keeps its contents without power. It stores firmware — the start-up program of a PC (BIOS/UEFI) and the program inside an irrigation controller or tractor engine control unit.

Type	How it is written	Can it be changed?
Mask ROM	Contents fixed by the chip maker during manufacture	No
PROM (programmable ROM)	Written once by the user with a programmer	No — one time only
EPROM (erasable PROM)	Written electrically; erased by ultraviolet light through a quartz window	Yes, after removing and erasing
EEPROM (electrically erasable PROM)	Written and erased electrically, byte by byte	Yes, in the circuit
Flash memory	A type of EEPROM erased in blocks	Yes — used for firmware, SSDs, pen drives and memory cards

Cache memory

Cache is a small amount of very fast SRAM between the CPU and main memory that holds copies of recently used data and instructions. Modern processors have several levels: **L1** (smallest and fastest, inside each core), **L2** and **L3** (larger, shared). Cache works because programs show **locality of reference** — they tend to reuse the same data soon (temporal locality) and to use data stored next to it (spatial locality).

- A **hit** means the data is found in the cache; a **miss** means it must be fetched from main memory.
- The **hit ratio** h = hits ÷ total accesses.

$$\text{Average access time } T = h \times T_c + (1 - h) \times (T_c + T_m)$$
$$T_c = \text{cache access time}, T_m = \text{main-memory access time}, h = \text{hit ratio}$$

Worked example — the value of a cache
h = 0.95, T_c = 1 ns, T_m = 60 ns.
T = 0.95 × 1 + 0.05 × (1 + 60) = 0.95 + 3.05 = 4.0 ns.
Without a cache every access would take 60 ns, so the cache makes memory access about 15 times faster.
(Some textbooks use the simpler form T = h·T_c + (1 - h)·T_m, which gives 3.95 ns here; state the form you use.)

Virtual memory

When programs need more memory than the RAM installed, the operating system uses part of the disk as an extension of RAM, moving blocks (**pages**) between the two as needed. This lets large programs run, but heavy use slows the computer because disks are far slower than RAM. Lecture 4 explains how the operating system manages this.

6. Secondary storage

Secondary storage is non-volatile and holds programs and data permanently, even when the computer is off.

Magnetic storage

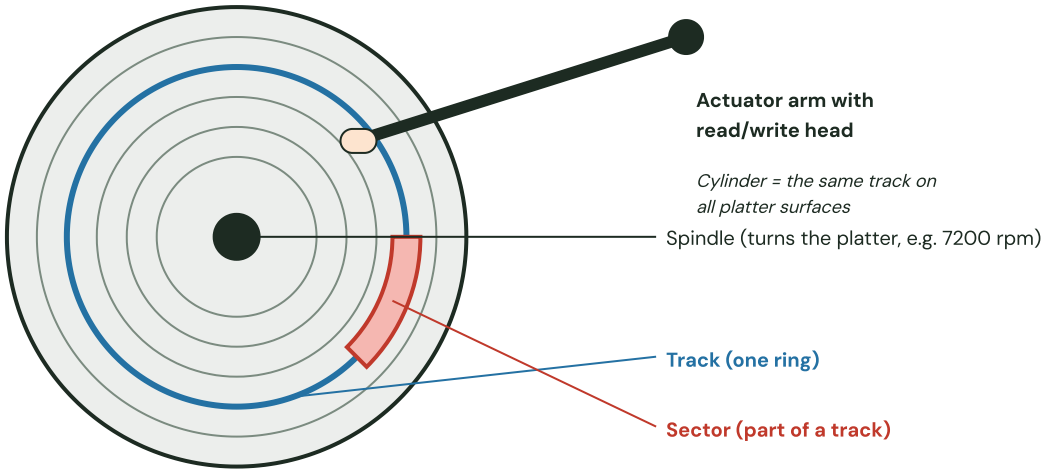


Figure 3. Top view of a hard-disk platter showing tracks, a sector and the read/write head. Original diagram.

- A **hard disk drive (HDD)** stores data magnetically on spinning **platters**. Each surface is divided into concentric **tracks**, each track into **sectors**, and the same track on all surfaces forms a **cylinder**. A **read/write head** on an actuator arm moves across the platters.
- **Access time = seek time + rotational latency + transfer time.** Seek time is the time to move the head to the track; rotational latency is the wait for the sector to come under the head.
- **Magnetic tape** stores data in sequence along a long tape. It is slow to search but cheap and durable, and is still used for large archives and backups.

> **Worked example — hard-disk access time**
> A disk spins at 7200 rpm = 7200 ÷ 60 = 120 revolutions per second.
> One revolution takes 1 ÷ 120 s = 8.33 ms; on average the sector is half a turn away, so rotational latency = 4.17 ms.
> With an average seek time of 9 ms and a transfer time of about 0.03 ms for a 4 KB block:
> Access time ≈ 9 + 4.17 + 0.03 ≈ 13.2 ms — over 200,000 times longer than a 60 ns RAM access.

Optical storage

Optical discs store data as microscopic pits and flat areas read by a laser. They are read in an optical drive and are used for software distribution and archives.

Disc	Typical capacity	Notes
CD (compact disc)	About 700 MB	Near-infrared laser
DVD	4.7 GB single-layer; 8.5 GB dual-layer	Red laser and smaller pits
Blu-ray Disc	25 GB single-layer; 50 GB dual-layer	Blue-violet laser; 5 to 10 times DVD capacity (Samsung India, 2020)

Solid-state (flash) storage

- **Solid-state drives (SSDs)** use flash memory with no moving parts. They are much faster than hard disks, silent and resistant to shock — a good choice for laptops used in the field.
- **USB flash drives (pen drives)** carry data between computers.
- **Memory cards** (SD and microSD) store data in cameras, drones, phones and data loggers.

SD card family (SD Association, n.d.)	Capacity	File system
SD (SDSC)	Up to 2 GB	FAT12/FAT16
SDHC	Over 2 GB up to 32 GB	FAT32
SDXC	Over 32 GB up to 2 TB	exFAT
SDUC	Over 2 TB up to 128 TB	exFAT

Before buying a card for a data logger or camera, check which family the device supports — an older device that reads SDHC may not read an SDXC card.

Cloud storage

Cloud storage keeps files on remote servers reached through the Internet (Lecture 7). It allows access from any device, sharing and off-site backup, but depends on connectivity and raises questions of privacy and cost.

Device	Access	Speed	Typical use on farms and in agri-offices
RAM	Random	Very fast	Running GIS, spreadsheets and models
SSD	Random	Fast	Field laptop; fast start-up
HDD	Direct	Moderate	Large archives of images and records in the office
Memory card / pen drive	Random	Moderate	Data loggers, drone and trail cameras, transfer of files
Optical disc	Direct	Slow	Distributing software and training videos
Magnetic tape	Sequential	Slow to search	Long-term archives of research data
Cloud	Over a network	Depends on connection	Sharing and backing up farm records

7. How much storage does farm data need?

Worked example 1 — a weather station
A station logs 6 parameters every 10 minutes, each stored as a 4-byte number.
Records per day = 24 × 6 = 144.
Bytes per day = 6 × 4 × 144 = 3,456 bytes.
Bytes per year = 3,456 × 365 = 1,261,440 bytes ≈ 1.26 MB (≈ 1.20 MiB).
Even a small memory card holds decades of such records.

Worked example 2 — a drone survey
A 20-megapixel camera stores 3 bytes per pixel (24-bit colour) before compression.
One image = 20,000,000 × 3 = 60,000,000 bytes = 60 MB.
A 500-image survey of a farm = 500 × 60 MB = 30,000 MB = 30 GB uncompressed.
Compression (JPEG) reduces this greatly, but image-based agriculture still needs gigabytes per survey — plan cards, drives and backups accordingly.

Protecting data

- **Back up regularly.** A widely used rule of thumb is **3-2-1**: keep three copies of important data, on two different kinds of media, with one copy off-site (for example, in the cloud).
- **Eject drives safely** before removing them, and keep media away from heat, moisture and strong magnets.
- **Erase data securely** before selling or discarding computers and cards, and send old devices to authorised e-waste collectors.

8. Key takeaways

- Computers store everything as bits; n bits give 2^n patterns, and 8 bits make a byte.
- Text is stored as character codes: ASCII (7-bit, 128 characters), ISCII and Unicode (UTF-8), which covers Indian scripts.
- Two unit conventions exist: binary (1 KB = 1024 bytes; IEC KiB) and decimal SI (1 kB = 1000 bytes). The newest SI prefixes are ronna (10^{27}) and quetta (10^{30}).
- The memory hierarchy trades speed and cost against capacity: registers → cache → RAM → SSD → HDD → archive.
- RAM (SRAM, DRAM) is volatile working memory; ROM (mask ROM, PROM, EPROM, EEPROM, flash) is non-volatile.
- Cache exploits locality of reference; virtual memory extends RAM using the disk.
- Secondary storage includes hard disks, SSDs, optical discs, memory cards, tape and the cloud.

9. Examination questions

Short-answer questions (2 marks)

Q1. Define a bit and a byte.

A bit is a single binary digit (0 or 1). A byte is a group of 8 bits.

Q2. How many different values can be stored in 10 bits?

2^{10} = 1024.

Q3. Convert 45 (decimal) to binary.

101101.

Q4. Distinguish between volatile and non-volatile memory with examples.

Volatile memory loses data without power (RAM); non-volatile memory keeps it (ROM, flash, hard disk).

Q5. Distinguish between SRAM and DRAM.

SRAM uses flip-flops, needs no refresh, is faster and costlier and is used for cache; DRAM uses capacitors, needs refreshing, is cheaper and denser and is used for main memory.

Q6. Expand EPROM and EEPROM. How is each erased?

Erasable programmable ROM (erased by ultraviolet light) and electrically erasable programmable ROM (erased electrically).

Q7. What is cache memory? Why does it work?

A small, fast SRAM memory between the CPU and main memory; it works because of locality of reference.

Q8. How many bytes are there in 1 GiB? In 1 GB (SI)?

1 GiB = 2^{30} = 1,073,741,824 bytes; 1 GB = 10^9 = 1,000,000,000 bytes.

Q9. Name the two newest SI prefixes and their values.

Ronna (R) = 10^{27} and quetta (Q) = 10^{30}, adopted in 2022.

Q10. What is the access time of a hard disk made up of?

Seek time + rotational latency + transfer time.

Long-answer questions (5–10 marks) — answer outlines

Q1. Explain the memory hierarchy with a diagram.

Draw Figure 2; describe each level with technology and typical access time; explain the speed–cost–capacity trade-off and why the hierarchy works (locality).

Q2. Classify computer memory and describe each type.

Primary (RAM: SRAM, DRAM; ROM: mask ROM, PROM, EPROM, EEPROM, flash; cache) and secondary (magnetic, optical, solid-state, cloud) — with uses.

Q3. Explain the units of memory, including binary and decimal prefixes.

Bit, nibble, byte, word; KB to YB with both conventions (table in Section 3); IEC binary prefixes; ronna and quetta; the 931 GB example.

Q4. Describe the working of a hard disk and calculate its access time.

Platters, tracks, sectors, cylinders, heads (Figure 3); access time components; the 7200 rpm example.

Q5. What is cache memory? Derive and use the average access time formula.

Levels L1–L3; locality; hit and miss; $T = h \cdot T_c + (1 - h)(T_c + T_m)$; worked example.

Q6. Estimate storage needs for a farm data system.

Weather-station and drone-survey calculations from Section 7; choice of media; the 3-2-1 backup rule.

10. Glossary

Term	Meaning
Address	A number identifying one memory location
ASCII	A 7-bit character code with 128 characters
Bit	A binary digit, 0 or 1
Byte	A group of 8 bits
Cache	Small, fast memory holding recently used data near the CPU
Cylinder	The same track on all surfaces of a hard disk
DRAM	Dynamic RAM, which stores bits as charges that must be refreshed
EEPROM	ROM that can be erased and rewritten electrically
Flash memory	EEPROM erased in blocks; used in SSDs and memory cards
Hit ratio	Fraction of memory accesses found in the cache
KiB, MiB, GiB	Binary units: 2^{10}, 2^{20}, 2^{30} bytes
Nibble	A group of 4 bits
RAM	Random-access memory — volatile read/write working memory
ROM	Read-only memory — non-volatile memory for firmware
Rotational latency	Waiting time for the right sector to reach the disk head
Sector	A part of a track on a disk
Seek time	Time to move the disk head to the right track
SRAM	Static RAM, built from flip-flops; used for cache
SSD	Solid-state drive, built from flash memory
Track	One concentric ring on a disk surface
Unicode	A character standard covering almost all the world's scripts
Virtual memory	Disk space used as an extension of RAM

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All diagrams are original. Explanations are written in original language; standard content on number systems, RAM and ROM types, cache and disks follows Patterson and Hennessy (2014), Rajaraman and Adabala (2015) and Sinha and Sinha (2004). Worked examples on farm data are illustrations prepared for this lecture.

Practice MCQs

Q1. One byte equals

(a) 4 bits (b) 8 bits
(c) 16 bits (d) 1024 bits

Q2. How many different values can 8 bits represent?

(a) 8 (b) 64
(c) 128 (d) 256

Q3. The binary equivalent of decimal 45 is

(a) 101011 (b) 101101
(c) 110101 (d) 100111

Q4. Which memory loses its contents when power is switched off?

(a) ROM (b) Flash
(c) RAM (d) Hard disk

Q5. Cache memory is usually made of

(a) DRAM (b) SRAM
(c) EPROM (d) magnetic tape

Q6. EPROM is erased by

(a) electric pulses (b) ultraviolet light
(c) magnetic field (d) heat

Q7. 1 GiB equals

(a) 10^9 bytes (b) 2^{30} bytes
(c) 2^{20} bytes (d) 10^{12} bytes

Q8. The SI prefix adopted in 2022 for 10^{30} is

(a) yotta (b) ronna
(c) quetta (d) zetta

Q9. With h = 0.9, T_c = 2 ns and T_m = 50 ns, the average access time T = h·T_c + (1 − h)(T_c + T_m) is

(a) 6.5 ns (b) 7.0 ns
(c) 5.2 ns (d) 50 ns

Q10. Average rotational latency of a 7200 rpm disk is about

(a) 1.4 ms (b) 4.17 ms
(c) 8.33 ms (d) 16.7 ms

Q11. An SDHC memory card can hold up to

(a) 2 GB (b) 32 GB
(c) 2 TB (d) 128 TB

Q12. Magnetic tape provides

(a) random access (b) sequential access
(c) direct access (d) cache access

Answers

Q1 — (b) 8 bits
Q2 — (d) 256
Q3 — (b) 101101
Q4 — (c) RAM
Q5 — (b) SRAM
Q6 — (b) ultraviolet light
Q7 — (b) 2^{30} bytes
Q8 — (c) quetta
Q9 — (b) 7.0 ns — *0.9 × 2 + 0.1 × 52 = 1.8 + 5.2 = 7.0 ns.*
Q10 — (b) 4.17 ms — *Half of 8.33 ms per revolution.*
Q11 — (b) 32 GB
Q12 — (b) sequential access